Again say we have n vectors in $\mathbb{R}^m$.

If $n < m$, these vectors cannot span $\mathbb{R}^m$.

Why? Well since there are n columns, the matrix has at most n pivots. And since there are m rows, at least one row has no pivot since $n < m$. Thus each row can't have a pivot so the columns can't span $\mathbb{R}^m$.

If $n > m$, do they span $\mathbb{R}^m$ then? Can't conclude that always unfortunately. What you can conclude is the set of vectors is then always linearly dependent.

Why? Say A is the matrix for these vectors (i.e. A's columns is this set of vectors). Linear dependence boils down to whether the homogeneous system $Ax = 0$ has a nontrivial solution. We showed that last time as setting a linear combination of those vectors to 0 is equivalent to that matrix equation.

And as we found earlier that system has a nontrivial solution exactly when A has a nonpivot column.

There are at most m pivots, and since $n > m$, that means at least one column has no pivot, proving the linear dependence.

So these two results, despite different settings and conclusions, use very similar logic to establish.

And together they tell us something quite important: the columns of an $m \times n$ real matrix can only possibly span $\mathbb{R}^m$ while being linear independent if $m = n$.

This is a symptom of why several chapters in this book (2, 3, 5) are focused solely on square matrices as those are very special. Many think we study square matrices because they are easier calculation-wise. And that's true, but overall it's deeper than that.

Since $n > m$ guarantees redundancy, let's consider $n < m$ to leave that issue more open. Say we have 50 vectors in $\mathbb{R}^{100}$. These can't span $\mathbb{R}^{100}$ so what do they span? If no redundancy, something in the shape of $\mathbb{R}^{50}$ perhaps? But if redundancy something smaller yes? Something like $\mathbb{R}^{49}$ or $\mathbb{R}^{40}$ or what?

How to figure that out exactly though?

Well if a vector is redundant, i.e. generated by the others, maybe you can throw it out since it doesn't really contribute anything? And then if remaining set is still linearly dependent can do same thing again?

That is an option, but a few things which make that potentially undesirable.

For one, as we stated last time even though a set of vectors may be redundant (i.e. linearly dependent), that doesn't mean every vector in the set is redundant, have to therefore check that carefully, can't arbitrarily pick.

And a vector which is redundant, if another vector is thrown out, maybe the original one is no longer redundant? So have to recheck that potentially.

So could be quite long. Maybe pivots have something to do with it since basic variables can always be expressed in terms of free variables (i.e they seem redundant). But again does that change once you throw one of them out?

In short, this requires more care to do properly, we'll return to it later.

1.7.5 Are columns of this matrix linearly independent?

$$\begin{bmatrix} 0 & -8 & 5 \\ 3 & -7 & 4 \\ -1 & 5 & -4 \\ 1 & -3 & 2 \end{bmatrix}$$

As stated earlier, boils down to whether each column has a pivot or not so put in REF to check.

Replace R2 by R2 - 3R4, R3 by R3 + R4 (we are sort of thinking of matrix being upside down with these)

$$\sim \begin{bmatrix} 0 & -8 & 5 \\ 0 & 2 & -2 \\ 0 & 2 & -2 \\ 1 & -3 & 2 \end{bmatrix}$$

Now move R4 to top and replace original R3 by R3 - R2:

$$\sim \begin{bmatrix} 1 & -3 & 2 \\ 0 & -8 & 5 \\ 0 & 2 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

Not quite REF, third row not fixed. But first two are, from which we discern that first two columns have pivots.

For last column, when we fix the third row via adding appropriate multiple of second row will last entry be 0?

Quite clearly it won't be so last column will have a pivot as well, proving this set is linearly independent.

1.8 Linear Transformations

A <u>transformation</u> T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a function from $\mathbb{R}^n$ to $\mathbb{R}^m$.

$\mathbb{R}^n$ is the <u>domain</u> of the function, the set of points where we define the function. And $\mathbb{R}^m$ is the <u>codomain</u> of the function, the set of *possible* outputs.

Do not confuse codomain and range! Range are the *actual* outputs only.

For instance $f: \mathbb{R} \dashrightarrow \mathbb{R}$ defined by $f(x) = x^2$

Here, as written, codomain is $\mathbb{R}$ but the actual range is $[0, \infty)$. So why not just make the codomain $[0, \infty)$. to avoid confusion? Well we could but then what about $g: \mathbb{R} \dashrightarrow \mathbb{R}$ defined by $g(x) = \frac{1}{x}$

Again, we can change the codomain of g to match its range. But g has a very different range than f. So doing that, it would by nature be in a different "category" than f due to different codomain then.

And many other functions which would also have very codomains if we chose to do that.

Point is we do not want to make a different category for each based on the new codomain. Easier usually to put them all in the same category based on same codomain $\mathbb{R}$ with the understanding that actual range of each might not be the entire codomain.

And sometimes we do that with domain also (though that is a little more situation dependent). But sometimes when we say $h: \mathbb{R} \dashrightarrow \mathbb{R}$ we don't mean that the domain of h is all of $\mathbb{R}$, it's some subset of $\mathbb{R}$, that for instance is what we meant above with g.

Once more we might have that convention to avoid having tons of different classes based on what the exact domain is, easier to put them all in one class with that understanding.

And again be careful not to confuse codomain and range as the question of whether for a particular function codomain and range are same is quite important! And assuming it's always true (which it certainly is not) will be a big problem!

A <u>linear transformation</u> $T: \mathbb{R}^n \dashrightarrow \mathbb{R}^m$ is a transformation which satisfies

(1) $T(u + v) = T(u) + T(v) \quad \forall \quad u, v \in \mathbb{R}^n$
(2) $T(cu) = cT(u) \quad \forall \quad c \in \mathbb{R} \ , \ \forall \quad u \in \mathbb{R}^n$

In essence linear transformation must preserve addition and scalar multiplication.

So in simplest case, $T: \mathbb{R} \dashrightarrow \mathbb{R}$ the type of function dealt with in single variable calculus, which of our well known functions is a linear transformation?

Based on name one might think linear functions, $f(x) = mx + b$, are the type that would be. But that's not quite true!

One key in any linear transformation is that we always have $T(0) = 0$. Why? Multiple ways to show that:

$$T(0) = T(0 + 0) = T(0) + T(0)$$

or

$$T(0) = T(c \cdot 0) = cT(0)$$

Cancelling in the first or taking c to be anything but 1 in the second, either way proves $T(0) = 0$.

Thus $f(x) = mx + b$ is not a linear transformation if $b \neq 0$ since then $f(0) = b$ is nonzero, contradicting what we just showed.

Hence from $\mathbb{R}$ to $\mathbb{R}$ actually the only linear transformations are of the type $f(x) = mx$

Could we think of that as multiplying by 1×1 matrix whose only entry is m? We could - that's an important prelude!

How to check if transformation is a linear transformation? Rather than showing both conditions hold, easier to check both at once via verifying

$$T(cu + v) = cT(u) + T(v)$$

holds $\forall\ c \in \mathbb{R}$, $\forall\ u, v \in \mathbb{R}^n$

Again though have to show this holds for all choices (not via checking one by one, some kind of proof). Key is you can't show it works for some specific c, u, v and then try to claim it always works - that's not safe or correct (this is a common mistake)!

Building off $f(x) = mx$, on any $\mathbb{R}^n$ to itself a common type of linear transformation is $T(u) = ru$ where r is any fixed real number. This is called a contraction or dilation depending on size of r.

Why is this a linear transformation? Well check the needed condition:

$$\begin{aligned} T(cu + v) &= r\,[cu + v] \\ &= c(ru) + rv \\ &= cT(u) + T(v) \end{aligned}$$

so that proves it is a linear transformation.

1.8.42 Say $T: \mathbb{R}^n \dashrightarrow \mathbb{R}^m$ linear transformation which maps a set of two linearly independent vectors onto a linearly dependent set. Show $T(x) = 0$ has a nontrivial solution.

Say $\{u, v\}$ are that linearly independent set in $\mathbb{R}^n$. Then

$$\{T(u), T(v)\}$$

is a linearly dependent set in $\mathbb{R}^m$ by what's given in problem. Thus by definition of linearly dependent there exist scalars c and d, not both 0, so that

$$cT(u) + dT(v) = 0$$

Then using properties of linear transformations we get

$$T(cu) + T(dv) = 0$$

---> $$T(cu + dv) = 0$$

Thus $cu + dv$ is our nontrivial solution to $T(x) = 0$. How do we know it is nontrivial? Well u, v are linearly independent so only way a linear combination of them, namely $cu + dv$ here, can be 0 is to have $c = d = 0$.

But we said earlier that not both are 0, so we are guaranteed that $cu + dv$ is nonzero and thereby is a nontrivial solution as we needed.

Although this is a proof, this is the sort of question I would consider reasonable level for a quiz/exam.

This indirectly also raises a question of a linear transformation "losing" flexibility as a linearly independent set maps onto a dependent one.

How to check if that happens in general?

And is it possible to gain flexibility instead of losing it in a linear transformation? I.e., map a linearly dependent set onto a linearly independent set?

If we express it as a matrix might we use matrix properties to help us answer things like this? We already mentioned this when we said $f(x) = mx$ could be thought of as multiplying by a 1×1 matrix.

These issues allude well to the next section.

1.9 Matrix for Linear Transformation

Say $T: \mathbb{R}^n \dashrightarrow \mathbb{R}^m$ is a linear transformation.

Can we perhaps express T in the form

$$T(x) = Ax$$

where A is a fixed $m \times n$ matrix? Again already mentioned doing that in case of $m = n = 1$ so trying to extend that idea.

Again if so, maybe understanding A better via matrix ideas will help us understand T better?

But ok two things to clarify with that. First off, is $T(x) = Ax$ always a linear transformation? And if so can we be sure every linear transformation can be expressed in this manner?

For the first part, say we have any $c \in \mathbb{R}$, $x, y \in \mathbb{R}^n$
Observe that

$$
\begin{aligned}
T(cx + y) &= A(cx + y) \\
&= A(cx) + Ay \qquad \text{(distributive law)} \\
&= c(Ax) + Ay \\
&= cT(x) + T(y)
\end{aligned}
$$

which proves T is linear transformation as this was shown to work for all c, x, y, not just some specific ones.

Again that is what must be done to really prove this is a linear transformation, and again this is kind of thing I would consider reasonable for quiz/exam.